



HABIB UNIVERSITY
Digital Logic and Design
EE-172 / CS-130 / CE-222
Fall 2025 (L2)

Quiz # 3

Student ID: _____

Student Name: _____

Date: October 30, 2025

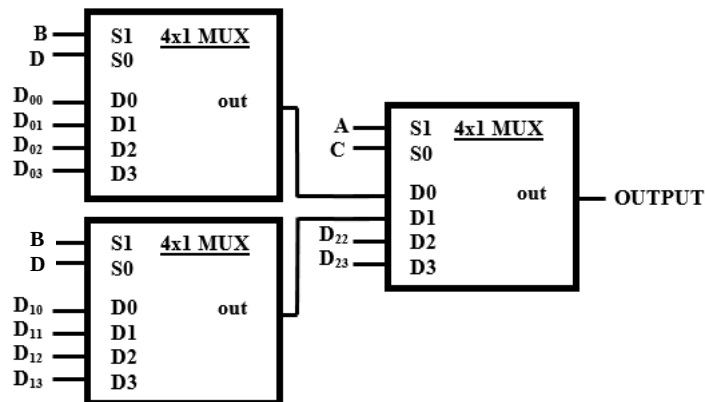
Time: 40 minutes

Max points: 24

Instructions:

- Smart watches, laptops, and similar electronics are strictly NOT allowed.**
- Answer sheets should contain all steps, working, explanations, and assumptions.**
- Attempt the quiz in the spaces provided for each question.
- You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

No.	Questions	Pts
1	Consider the given figure. Three 4x1 multiplexers are interconnected to form an Nx1 multiplexer. - What is the value of N? - What should be value of ABCD to route data from: - D₁₂ to the output? - D₂₂ to the output?	1 4



- N=10. It is the number of inputs. Here a total of 10 external inputs are there.
- To route D₁₂ to the output: ABCD = 0110
To route D₂₂ to the output: ABCD = 1X0X

2 Using the given truth table, design the circuit of a 4-to-2 priority encoder.

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Inputs				Outputs		
D ₀	D ₁	D ₂	D ₃	V	A	B
0	0	0	0	0	X	X
1	0	0	0	1	0	0
X	1	0	0	1	0	1
X	X	1	0	1	1	0
X	X	X	1	1	1	1

Equations:

$$A = D_2 + D_3$$

$$B = D_1 \overline{D_2} \overline{D_3} + D_3 = D_1 \overline{D_2} + D_3$$

$$V = D_0 + D_1 + D_2 + D_3$$

Draw circuit.

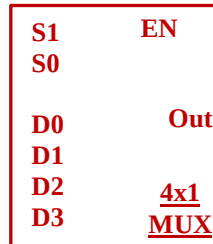
An 8 x 1 multiplexer has 8 inputs, 3 select lines and 1 output.

Let the 8 inputs be d0, d1, d2, d3, d4, d5, d6 and d7,

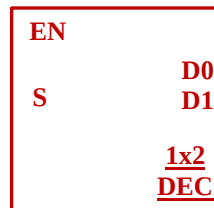
and the 3 select lines be A (MSB), B and C.

F represents the output.

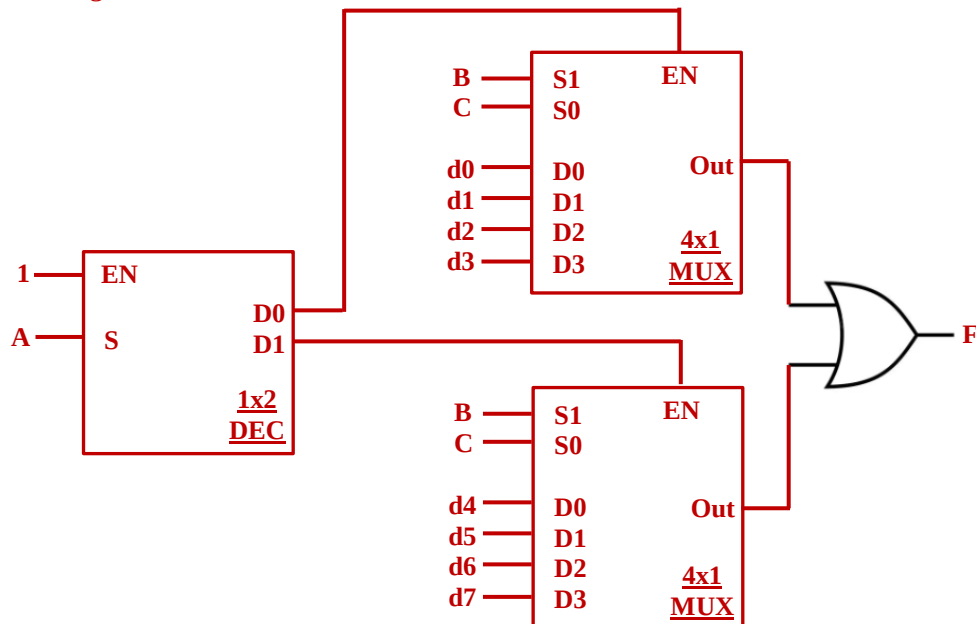
Consider the following block diagram for 4x1 multiplexer, where S1 (MSB) and S0 are select lines, D0, D1, D2 and D3 are inputs, Out represents the output, and EN is the enable input.



Consider the following block diagram for 1x2 decoder, where S is the input, D0 and D1 are the outputs and EN is the enable input.



Cascading:



Let A and B be the two input operand bits and C_{in} be the input carry. S and C_{out} respectively represent the sum and output carry.

Truth Table:

A	B	C_{in}	C_{out}	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Equations:

$$\begin{aligned}
 S(A, B, C_{in}) &= \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in} \\
 &= \bar{A}(\bar{B}C_{in} + B\bar{C}_{in}) + A(\bar{B}\bar{C}_{in} + BC_{in}) \\
 &= \bar{A}(B \oplus C_{in}) + A(B \odot C_{in}) \\
 &= \bar{A}(B \oplus C_{in}) + A(B \oplus C_{in}) \\
 &= \bar{A} \oplus B \oplus C_{in}
 \end{aligned}$$

$$\begin{aligned}
 C_{out}(A, B, C_{in}) &= \bar{A}BC_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in} \\
 &= (\bar{A}B + A\bar{B})C_{in} + AB(\bar{C}_{in} + C_{in}) \\
 &= (A \oplus B)C_{in} + AB(1) \\
 &= (A \oplus B)C_{in} + AB
 \end{aligned}$$

Draw circuit.